

WAT.ai

weatherloo onboarding

Prepared by Cindy & Ayaan

Agenda

- Introduction
- Problem Scope
- Timeline
- Communication
- Form
- Action Items



Introduction



Ayaan Rezwan (he/him)

Program + Term: 1B Biomedical Engineering

Area of interest: Computational Genomics +
Machine Learning

Fun fact: I work with radioactive & toxic materials!!

Introduction



Cindy Li (she/her)

Program + Term: 4A Computer Science

Area of interest: Machine Learning + Agent Infra

Fun fact: I interned for a year as an AI Meteorologist, which is how I got into this topic.

Goal 1: publish the best weather forecast for Waterloo

- Deliverable 1: Public app/website for the community to use
- Deliverable 2: Paper published on arXiv

Goal 2: allow anybody to replicate our method for anywhere in the world

- Deliverable 3: Website hosting an automated ML workflow where users can select any region in the world and we automate the pre-processing/training/running
- (Deliverable 4): Technical write-up of our autoML workflow

How are we going to be the best?

Post-processing and downscaling.

One of the major problems of local weather forecasts is that global forecasting models do not capture regional trends, leading to predictable biases. We want to build an ML-based post-processing layer on top of existing products to correct these biases.

The other problem is that global forecasting models have a relatively low resolution. We want to downscale forecasts to capture finer details.

Timeline (June)



Main Platforms



Values

Be friends!!!

Communicate openly.

Passion.

Task: Benchmarking Site

Motivation: we must understand what we need to beat + will introduce you to the key terminology and concepts relating to weather forecast evaluation.

See [/benchmarking-site](#) and GitHub Issues. Talk to Cindy if you have questions.

Task: Lit Review

Motivation: we must understand the ML weather forecasting techniques to create our own + will prepare us for model training ahead.

See /lit-review. Talk to Ayaan if you have questions.

What TPMs are up to:

- Connecting with professors and researchers for ideas and guidance on the project
- Contacting potential industry sponsors
- Lit review
- Answering questions
- Planning socials

Onboarding Form/WATCloud Access

(on Discord)

Action Items

1 Day → Onboarding form, GitHub, WATCloud access

1 Week → Lit Review & Benchmarking

1 Month → Start developing/training architecture

Any questions?



Email

c52li@uwaterloo.ca | arezwan@uwaterloo.ca

Discord

@cindehaa | @ayaonic

Take-Home Demos!!